

■# PAPER<i>348 — Stephan's Quintet Shock Ridge: F</i>U<i>Bi</i>i with KE Density and LENR Energy Coupling

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 96 **Source:** gokshare31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF FUBi_i for an intergalactic shock ridge with LENR coupling **Author:** Daniel T. Murphy

<!-- UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $M_{UQFF} = 1.43e1 \text{ TeV}$ -->

Abstract

The complete UQFF buoyancy-unified force FUBii is computed for the Stephan's Quintet compact group intergalactic shock ridge. The 1500 km/s relative velocity of the NGC 7318b intruder galaxy generates a kinetic energy density $KE_{den} = \frac{1}{2}\rho \cdot \Delta v^2$, which couples to the UQFF vacuum field via FLEENR (Low Energy Nuclear Reaction force component). The shock ridge lies at $x_2 = 290 \text{ Mly}$ and yields $FUBii \approx -8.32 \times 10^{21} \text{ N}$.

2. Core Physics

2.1 UQFF Buoyancy-Unified Force

$$F_{U_Bi_i} \approx -8.32 \times 10^{21} \text{ N}$$

2.2 Shock Kinetic Energy Density

$$KE_{\text{den}} = \frac{1}{2} \rho_{\text{IGM}} \cdot \Delta v^2 = \frac{1}{2} \rho_{\text{IGM}} \cdot (1500 \times 10^3 \text{ m/s})^2$$

where $\rho_{\text{IGM}} \approx 10^{-2} \text{ kg/m}^3$ (intragroup medium density at $z \approx 0.021$).

$$KE_{\text{den}} \approx \frac{1}{2} \times 10^{-2} \times (1.5 \times 10^6)^2 = 1.125 \times 10^{-14} \text{ J/m}^3$$

2.3 LENR Energy Coupling (Kozima/FLEENR)

The UQFF includes a Low Energy Nuclear Reaction force component:

$$E_{\text{FLEENR}} = E_{\text{Kozima}} \cdot \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} \cdot [SSq]$$

where E_{Kozima} represents nuclear binding energy release in dense shock-compressed plasma.

2.4 Cross-System Separation

$$x_2 = 290 \text{ Mly} = 2.90 \times 10^{23} \text{ m}$$

Distance from observer to the Stephan's Quintet shock ridge (Hickson Compact Group HCG 92).

3. Key Values

Quantity	Formula	Value
Δv	Relative shock velocity	1500 km/s
KE_den	$\frac{1}{2}\rho\cdot\Delta v^2$	$\sim 10^{-1}\blacksquare\text{ J/m}^3$
FUBi_i	UQFF full 5-eq	$-8.32\times 10^{21}\blacksquare\text{ N}$
x_2	Distance (observer)	290 Mly
E_FLENR	LENR coupling	$\text{Kozima} \times [\text{SSq}] \times \rho_{\text{ratio}}$
ρ_{IGM}	Intragroup medium	$\sim 10^{-2}\blacksquare\text{ kg/m}^3$

4. Physical Significance

Stephan's Quintet is the most famous compact group collision, extensively studied with JWST Cycle 1 data (2022–2025). The UQFF shock ridge model is unique in connecting the kinetic energy density of the 1500 km/s shock to LENR vacuum coupling — a novel physical mechanism not present in standard hydrodynamic models. The FLEENR term represents the possibility that ultra-high-density shock fronts ($>10\blacksquare\times$ ambient) may trigger sub-threshold nuclear reactions mediated by vacuum buoyancy.

The $x_2 = 290\text{ Mly}$ cosmic baseline sets the scale for UQFF long-range vacuum coherence tests: the $FUBii \approx -8.32\times 10^{21}\blacksquare\text{ N}$ over 290 Mly suggests that UQFF maintains coherent force coupling at intergalactic baselines.

5. Deduplication Note

vs. PAPER346, PAPER347: Those are AGN jet systems; this paper is an intergalactic shock ridge (no central BH jet).
****vs. PAPER_351 (ASASSN-14li):**** Both include F_Kozima; ASASSN-14li is a TDE (stellar scale), Stephan's Quintet is intergalactic.

6. Classification

Physics Territory: FIRST UQFF intergalactic shock ridge with KE_den and LENR coupling **Scale:** Intergalactic (290 Mly) **CP Implementation:** StephansQuintetShockRidgeFUBiCalculator (CondensedPhysics3.py, Session 96)